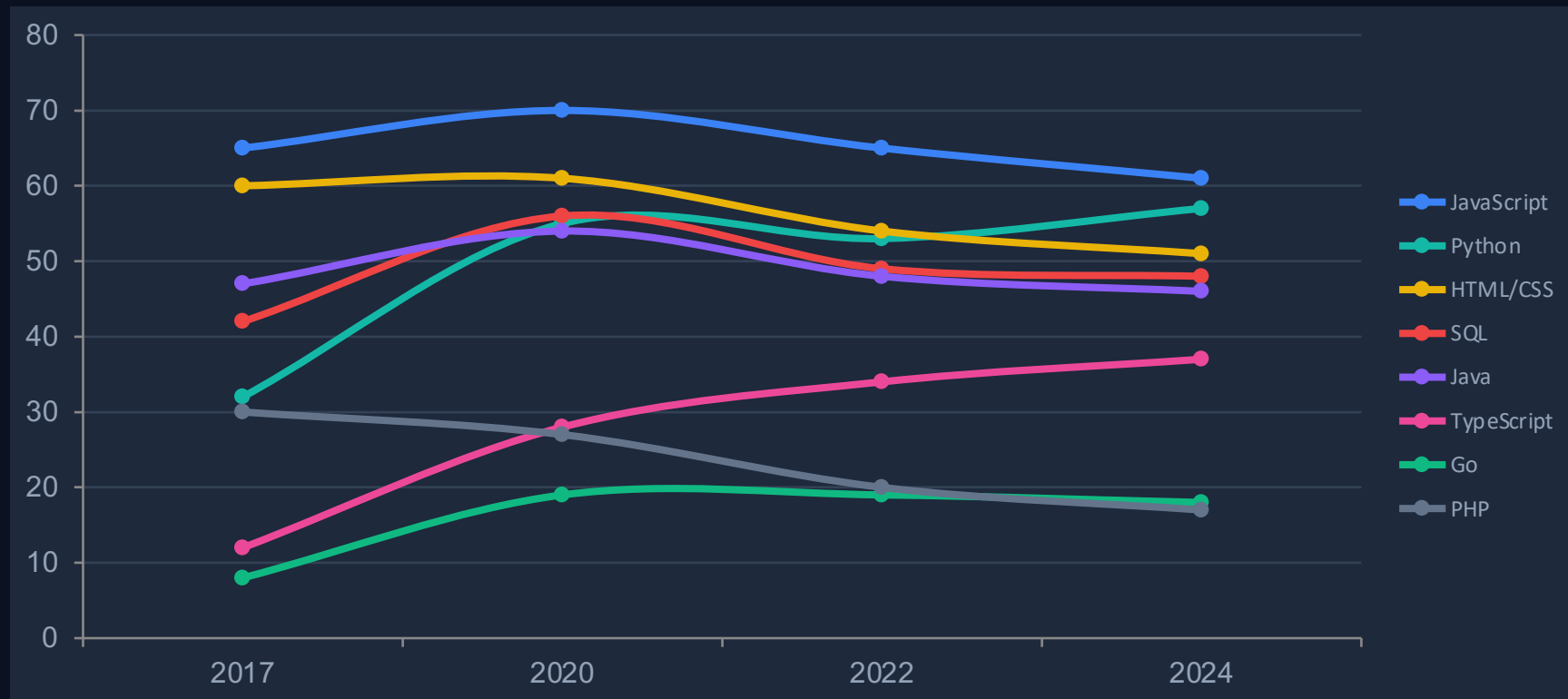


State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

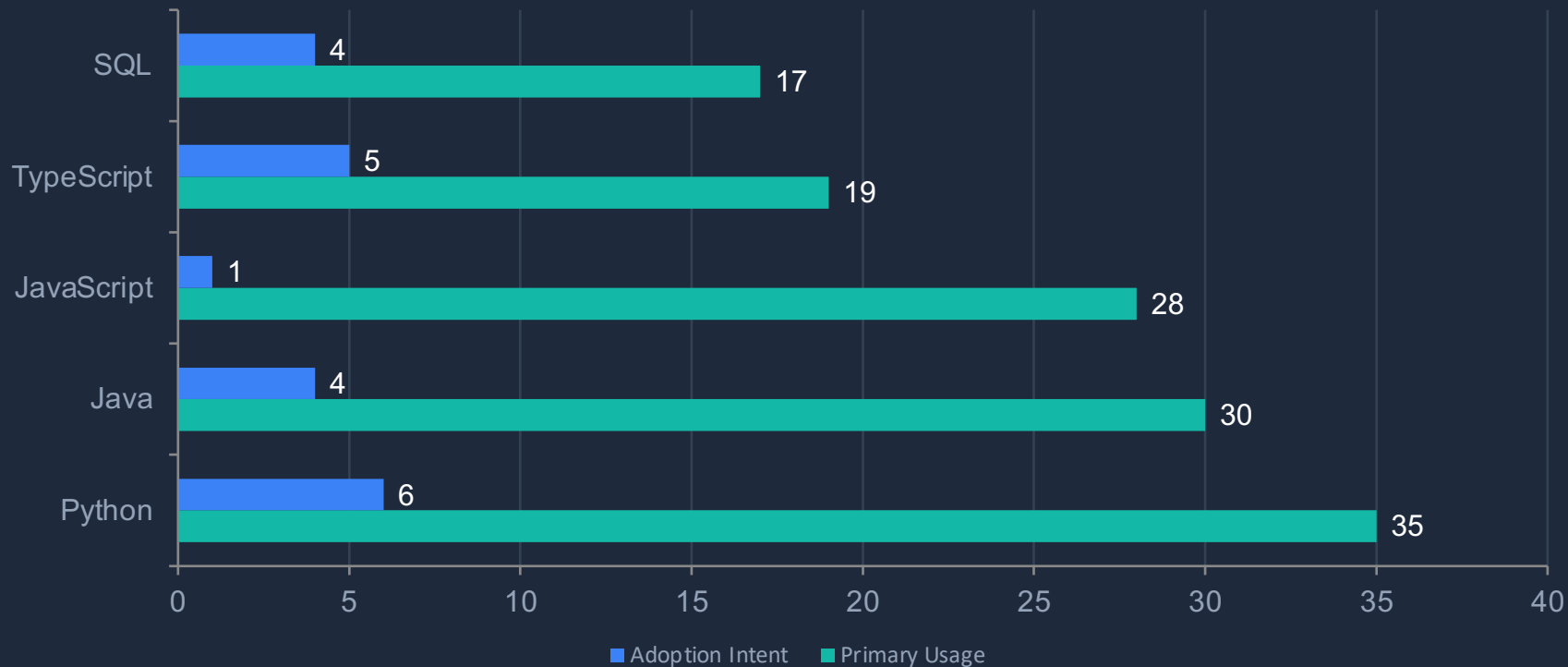
JetBrains' annual survey of 23,262 developers worldwide.

The most comprehensive snapshot of the developer community, covering languages, tools, AI attitudes, and career trends.

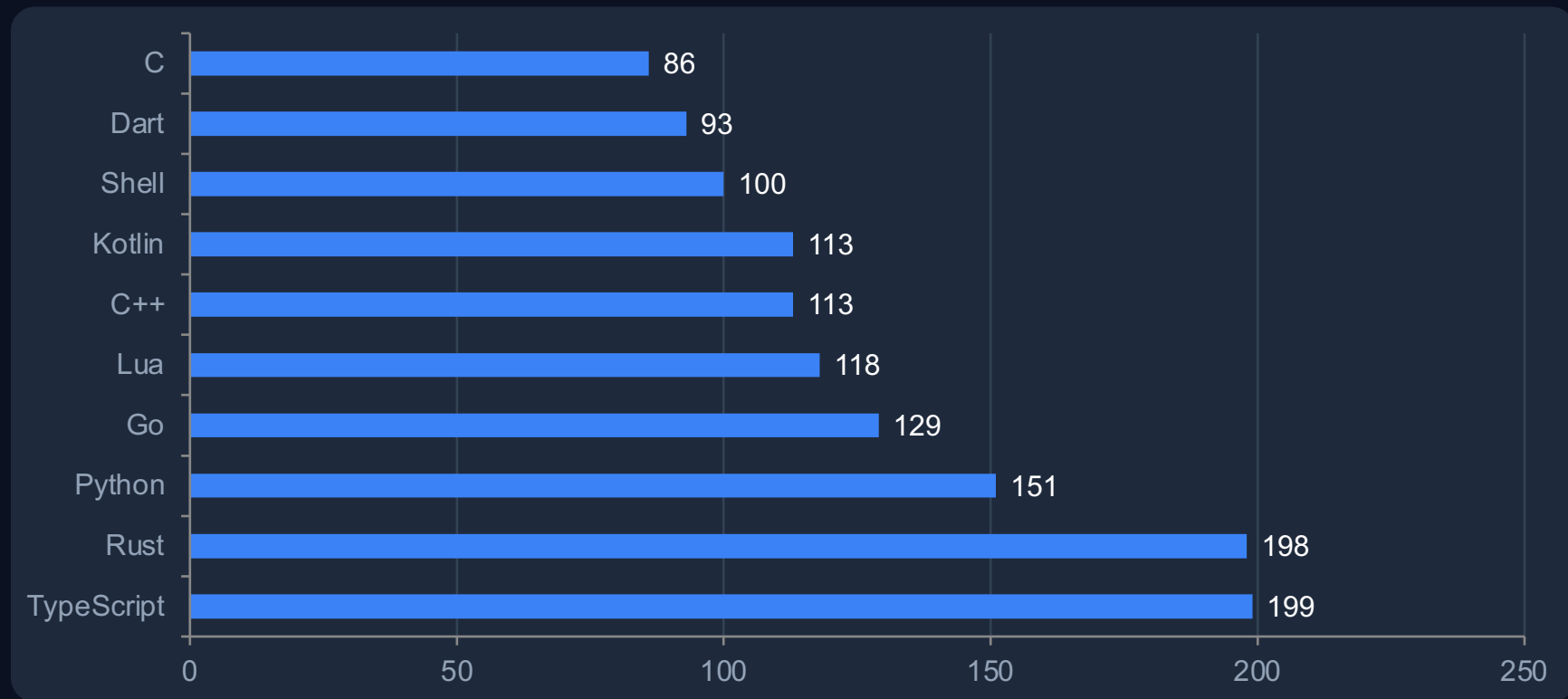
Python Closes the Gap: 8 Years of Language Usage Shifts



Python Is Now the #1 Primary Language at 35%



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

AI / Machine Learning

Python (31%)

Data Science

Python (28%)

User Interface

TypeScript / JS (62% / 58%)

APIs & Services

Java (49%)

Application Logic

Java (55%)

System Tools & Components

C++ (27%)

What This Means for Your Technology Strategy in 2025

- 1 Invest in Python for AI/ML teams — it is the undisputed leader in this space.
- 2 TypeScript is the future of frontend and full-stack development.
- 3 Rust and Go are the growth plays for infrastructure and systems programming.
- 4 PHP and Ruby hiring pools are shrinking as they lose ground to modern alternatives.
- 5 Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy.